Extreme Weather and Public Health

Assessing the impact of extreme weather on excess deaths

Tom Rose and Sunit Patil

Our data

Health data:

- The UK has publicly available public health statistics for a large population (~60-70 million) dating back to 1981, published online by the ONS.

- What's in the statistics? Weekly deaths in the UK dating back to 1981, with regional and age breakdowns.

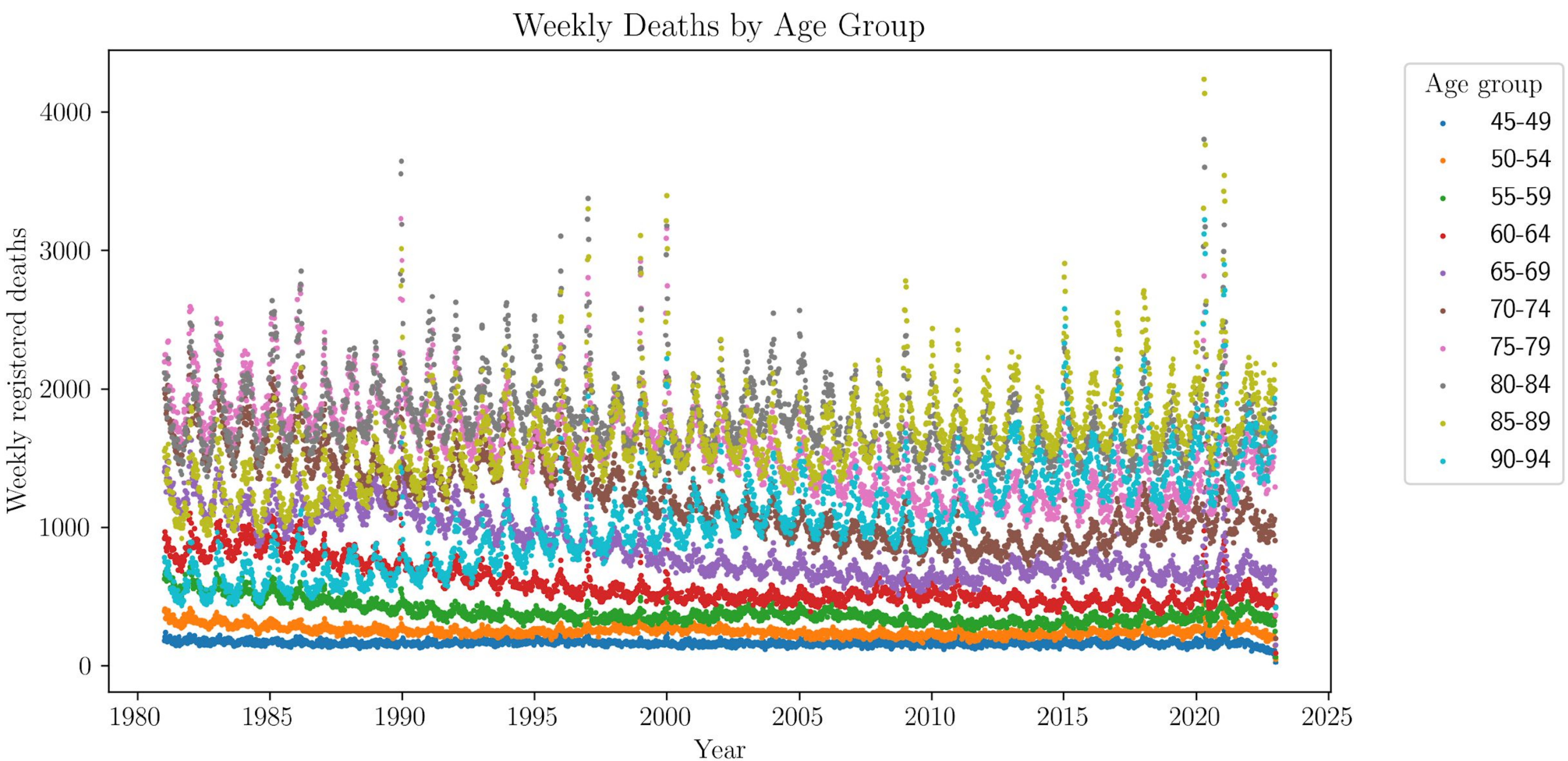
Weather data:

- We combined these with daily historical weather data to determine the if there is an impact from extreme weather events (from visualcrossing.com).

- We dropped 2020 onwards due to the difficult to quantify effects of Covid-19

A glance at the data

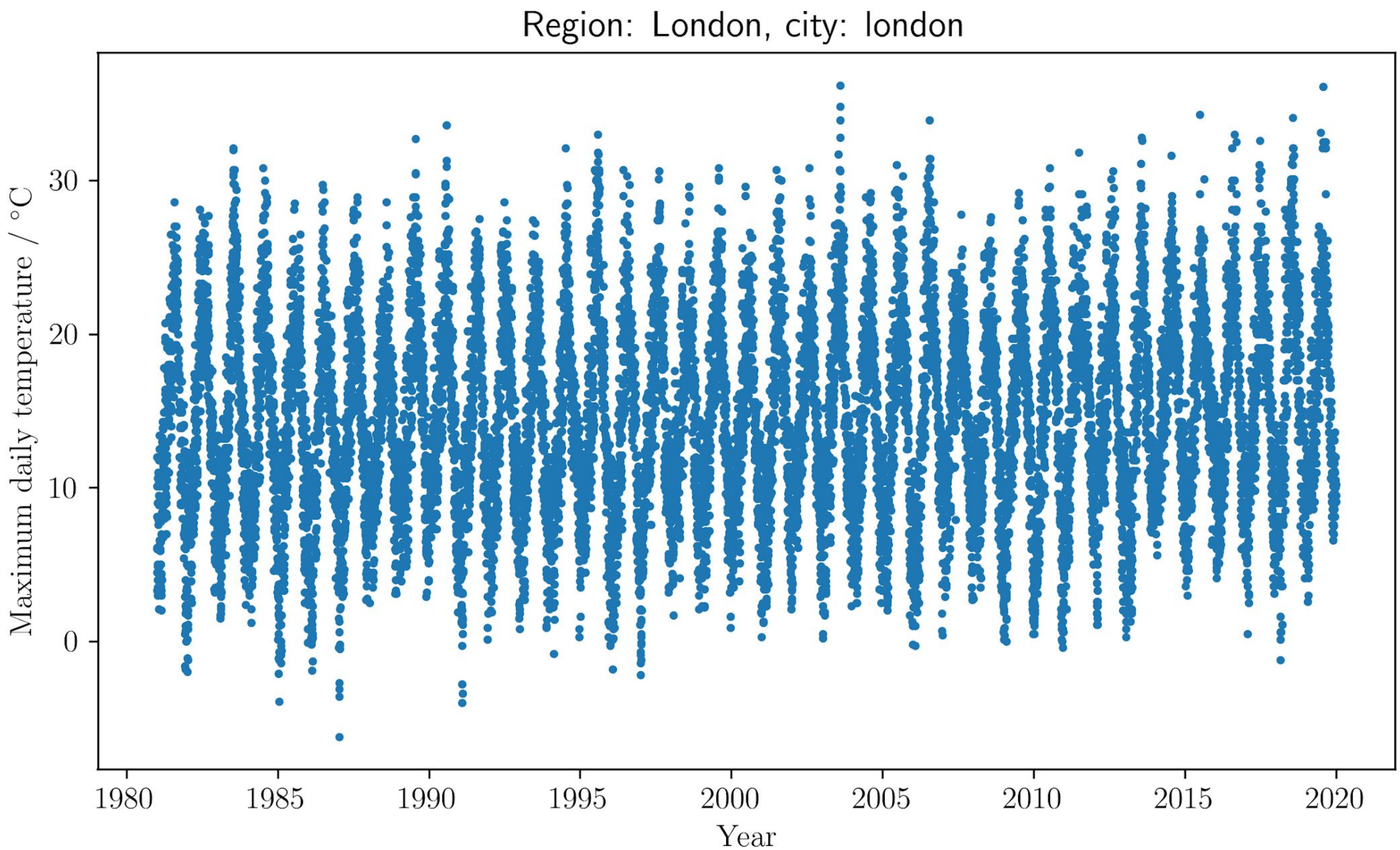
Weekly deaths over time



Plot loc: /Data/PreliminaryDataExploration/deaths_vs_time_by_age/

A glance at the data

Temperature over time



Plot loc: /Data/PreliminaryDataExploration/temperature_vs_time

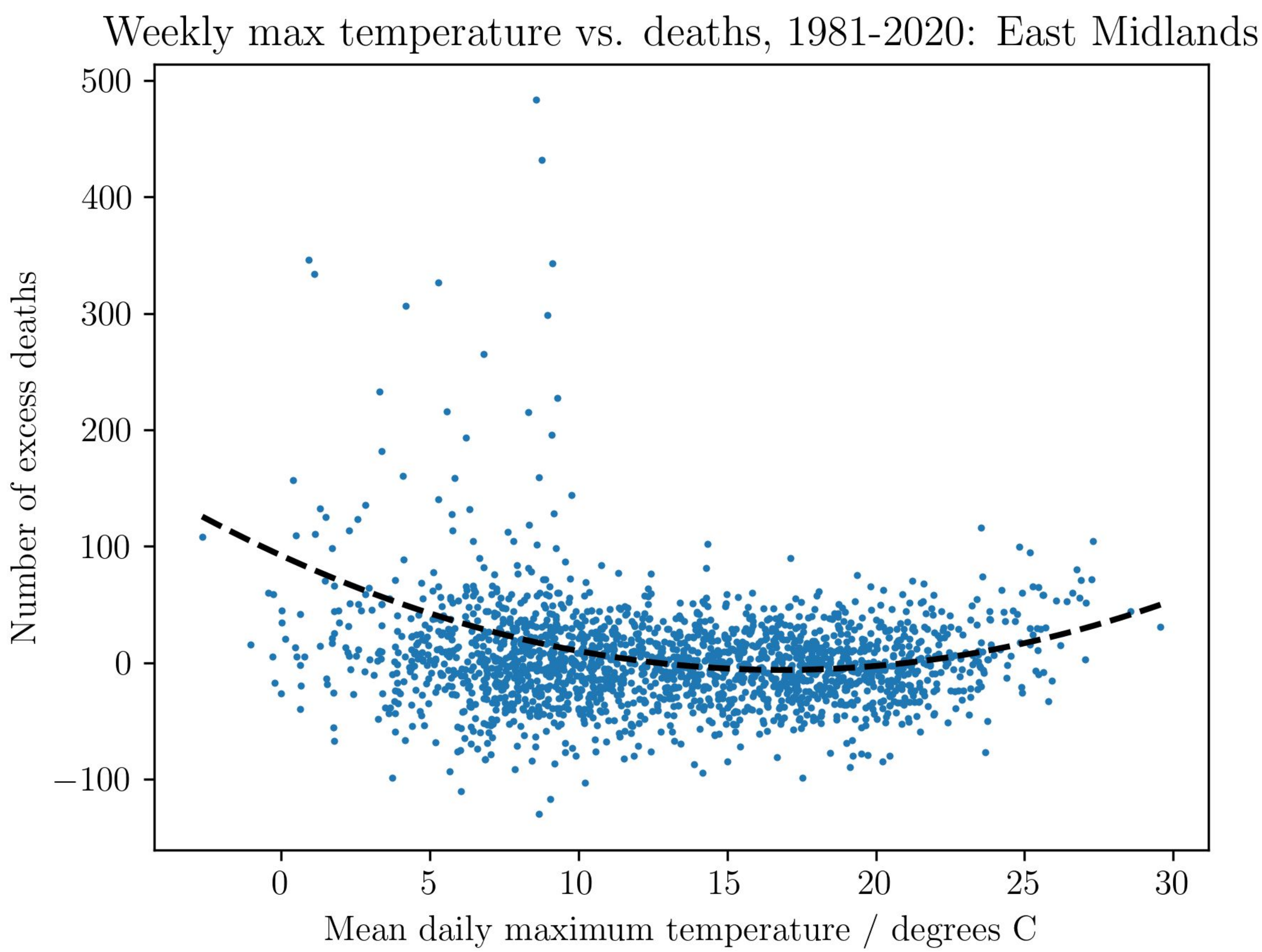
Feature Engineering

Our data contains many features. Throughout our initial analysis, we found some useful and others less useful.

- Weather data:
    - The most significant features we found to be correlated with mortality were temperature, precipitation and snowfall.
    - Features such as cloud cover and wind direction had no discernible effect and were dropped.
- Health data:
    - Contains many useful features, such as age and regional breakdowns.
    - To maintain good sample sizes, we chose a breakdown by region rather than age. With more time, an analysis of both may be useful.

The relationship between temperature and deaths after removing the seasonal trends

A simple polynomial model:



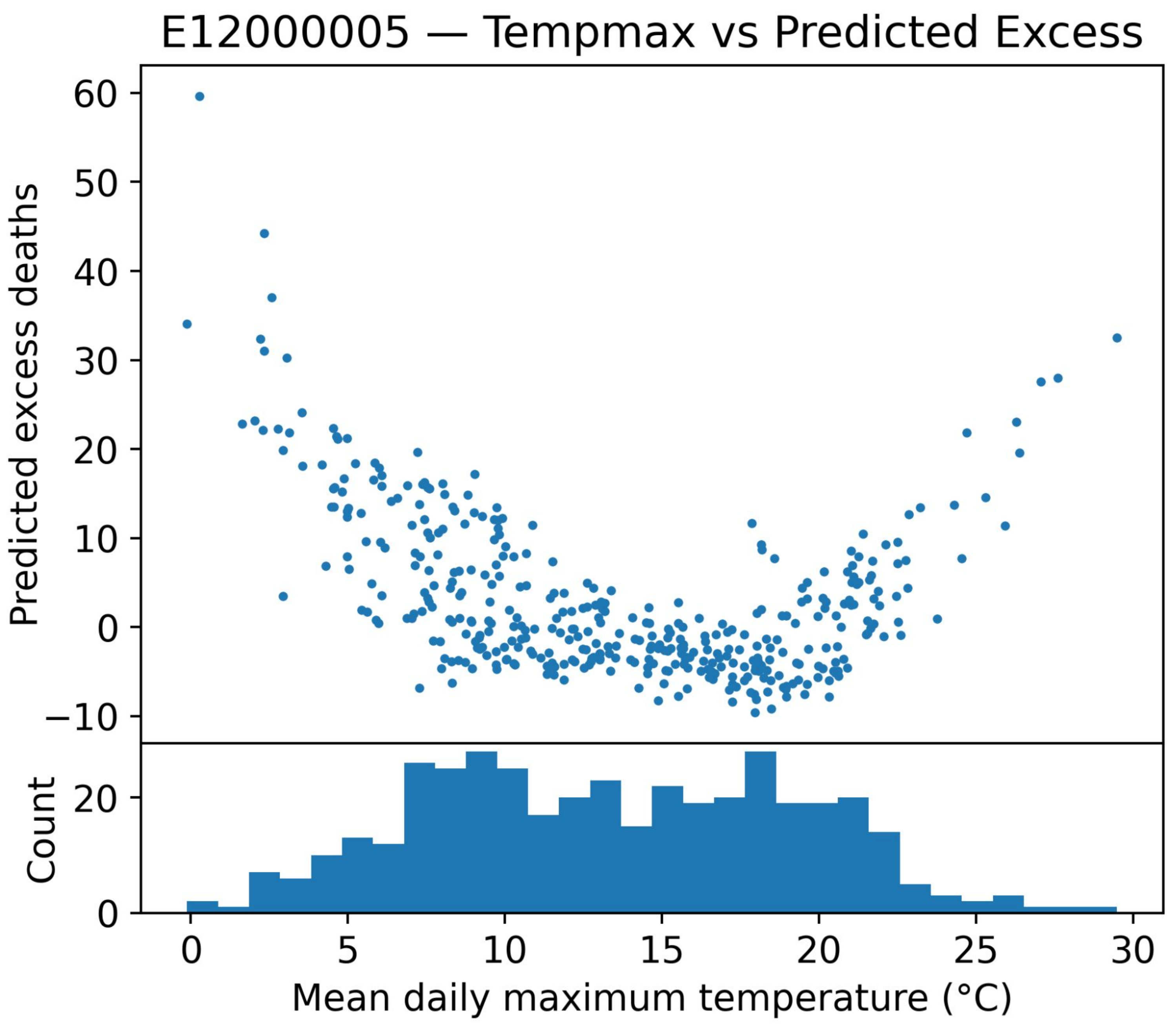
A second order polynomial linking the week's mean temperature to excess deaths.

There is a clear effect from the extreme hot and cold temperatures.

Plot loc: /Data/BasicPolynomialModeling/deaths_vs_temperature/region_breakdown

A CNN model

The trained model takes weather data (temperature, precipitation etc.) and predicts the effect on the corresponding period's deaths.

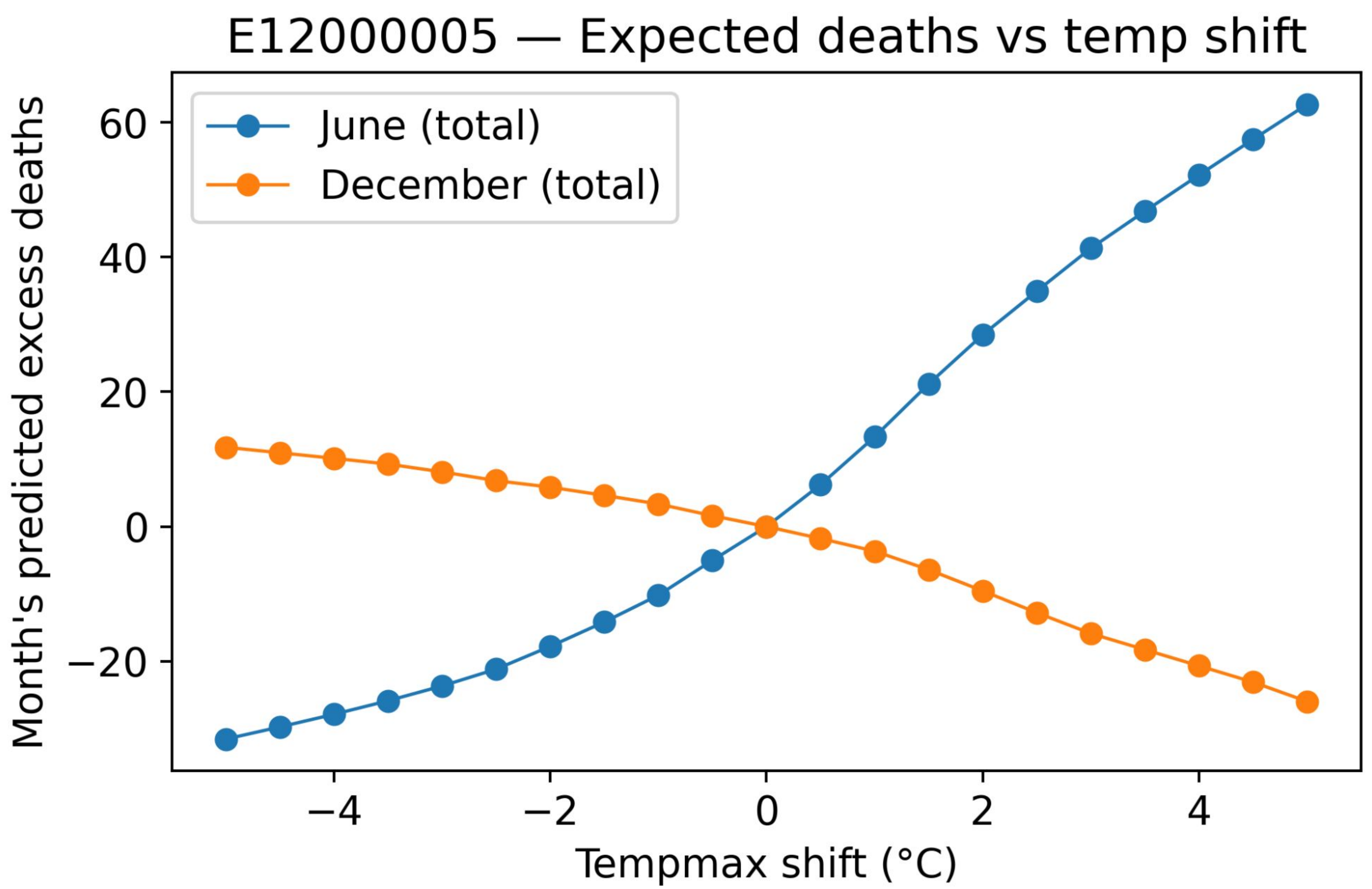


Plot loc: /Data/PyTorchModellingAndPredictions/CNN_Modelling_deseasonalised/region_splits/E12000005

Model Usage

Predicting upcoming excess deaths, an example

Input: weather data —> Output: expected excess deaths



Example: if the average temperature in June is 5°C above average, roughly 60 excess deaths are expected in this region.

Plot loc: /Data/PyTorchModellingAndPredictions/CNN_Modelling_deseasonalised/region_splits/E12000005

Modeling Temperature-Mortality using Distributed Lag Non-linear Model (DLNM)

We use the public health research standard DLNM model based on the non-linear, lagged effect of temperature, while controlling for long-term trend, annual seasonality, and day-of-week effects.

Model Outputs:

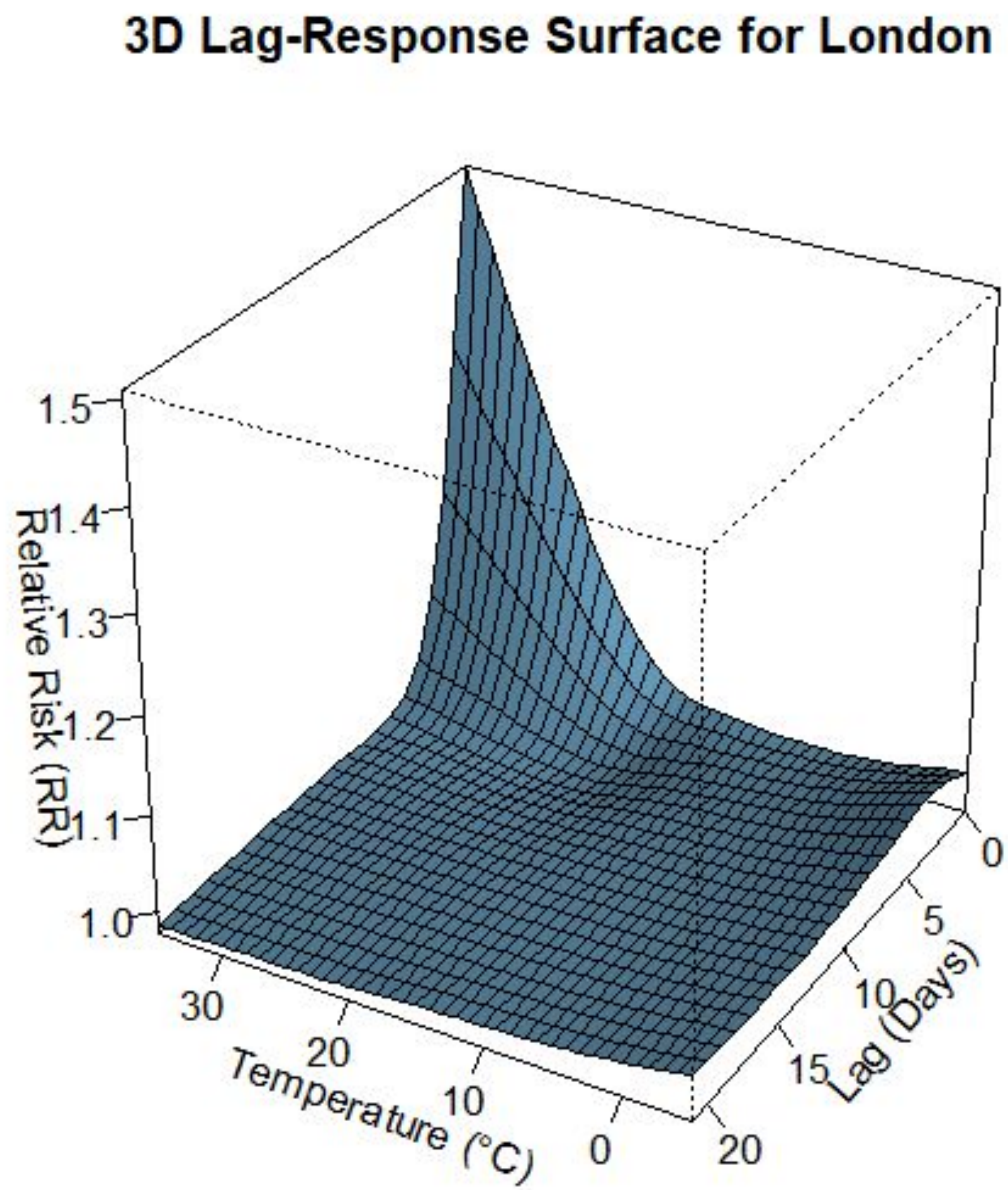
MMT (Minimum Mortality Temperature)

- The single "safest" temperature where mortality risk is at its absolute lowest. This is the bottom of the "U" or "J" curve.

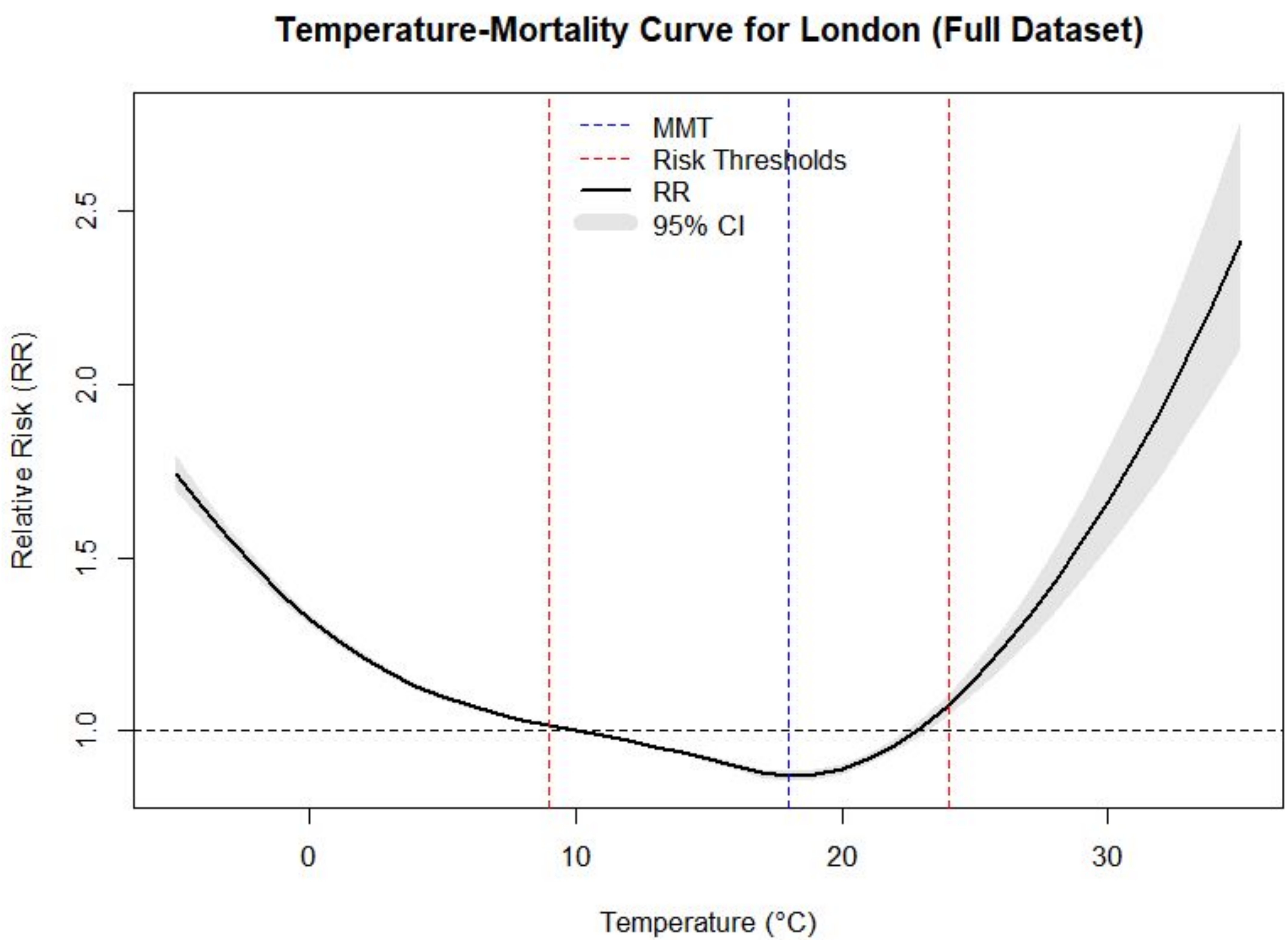
Risk Thresholds (Cold & Heat Cutoffs)

- The specific temperatures (e.g., 9°C or 24°C) where the mortality risk (given by Relative Risk RR) becomes statistically significant (RR > 1, with 95% confidence).

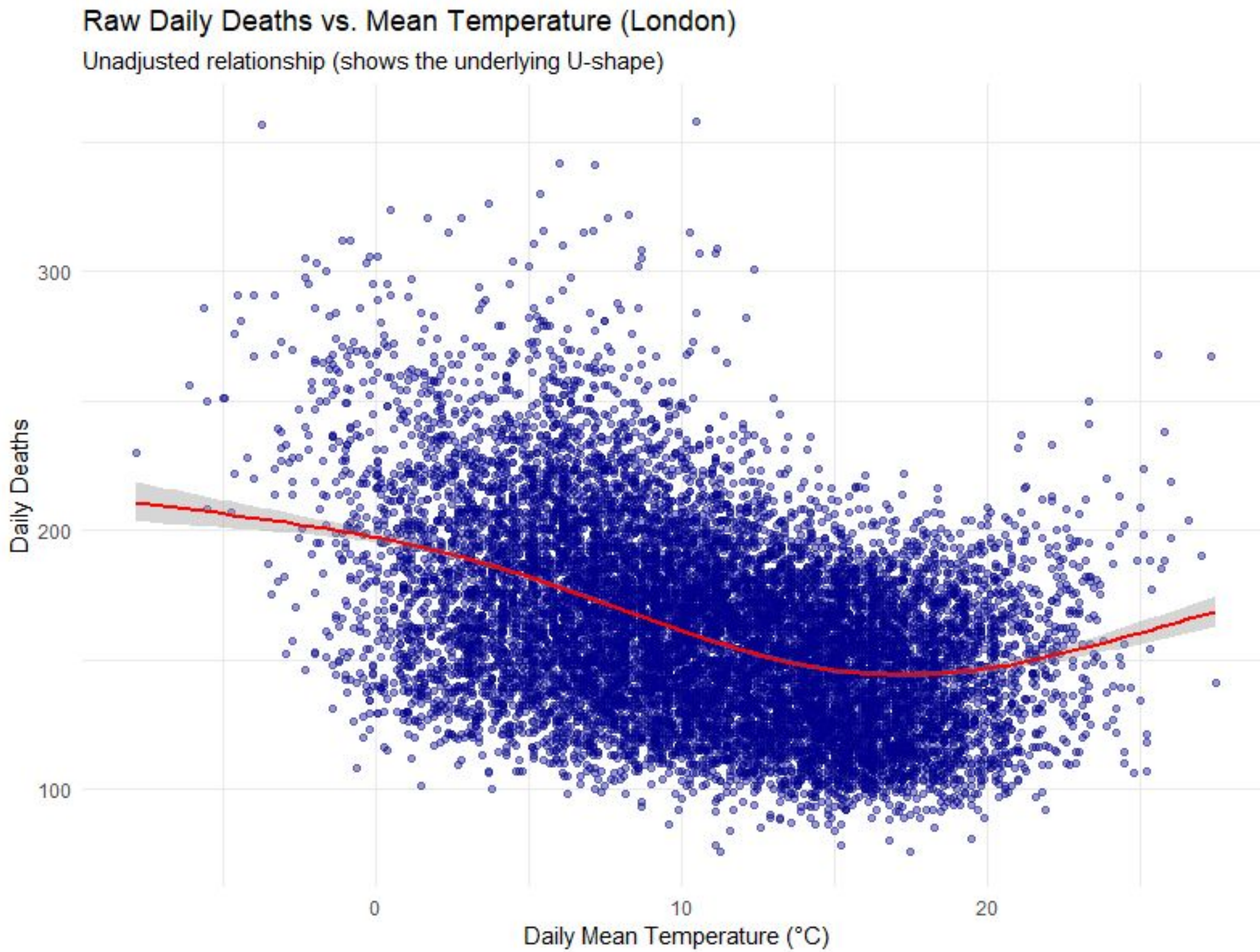
DLNM generated 3D Lag - Temperature - RR



2D Cross-section



Daily Deaths vs Temperature Scatter Plot



Results from different regions

Region	MMT (Safest Temp.)	Cold Threshold	Heat Threshold	Curve Shape
London	18°C	9°C	24°C	U-Shape
South East	19°C	9°C	26°C	U-Shape
West Midlands	19°C	9°C	27°C	U-Shape
East of England	19°C	9°C	29°C	U-Shape
East Midlands	19°C	9°C	30°C	U-Shape
Yorkshire & Humber	20°C	9°C	NA	J-Shape
South West	20°C	9°C	NA	J-Shape
North West	21°C	9°C	NA	J-Shape
North East	27°C	9°C	NA	J-Shape

KPI follow-up

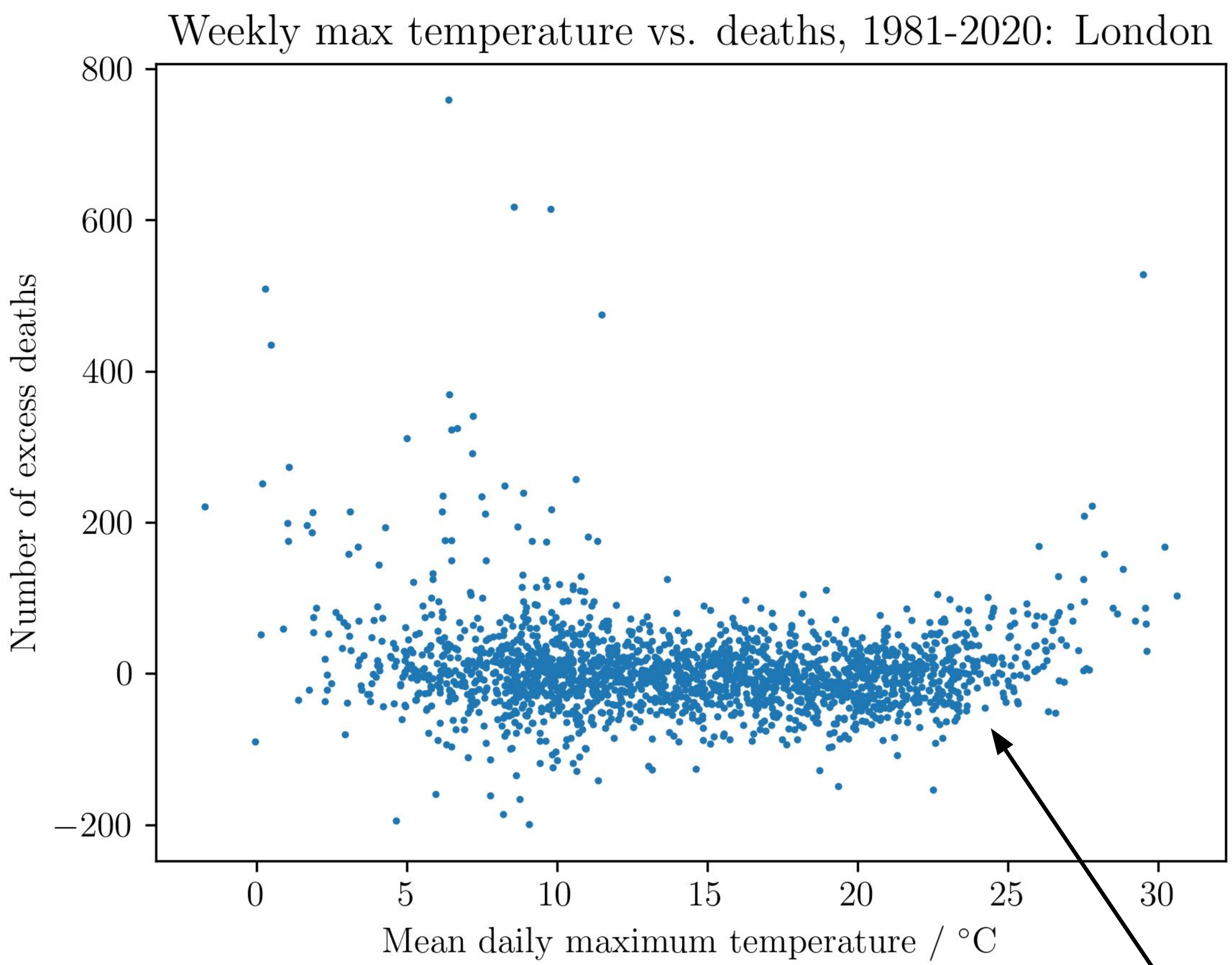
- KPI 1: Accuracy on unseen data.
- Achieved? Yes, despite high noise levels in the data, our models were able to pick up the underlying trends in both the training and test data.

- KPI 2: Consistency of effect direction/magnitude across regions.
- Achieved? Yes, we observe a similar effect across regions.

- KPI 3: Clarity/usefulness of stakeholder-facing summaries & indicators.
- Achieved? Partly. Due to the high noise and weak signal, the models will be useful to some but not all stakeholders e.g. to epidemiologists but not for surge planning, except in extreme cases.

- KPI 4: Reproducibility of data processing and modeling pipeline.
- Achieved? Yes, our findings were reproducible with a variety of different models and model variants

The relationship between temperature and deaths after removing the seasonal trends



Each point represents one week

Plot loc: /Data/PreliminaryDataExploration/temperature_vs_deaths/sub_moving_avg

Modeling Temperature-Mortality using Distributed Lag Non-linear Model (DLNM)

We use the public health research standard DLNM model based on the non-linear, lagged effect of temperature, while controlling for long-term trend, annual seasonality, and day-of-week effects.

Model Outputs:

MMT (Minimum Mortality Temperature)

- The single "safest" temperature where mortality risk is at its absolute lowest. This is the bottom of the "U" or "J" curve.

Risk Thresholds (Cold & Heat Cutoffs)

- The specific temperatures (e.g., 9°C or 24°C) where the mortality risk (given by Relative Risk RR) becomes statistically significant (RR > 1, with 95% confidence).

R-squared (R^2)

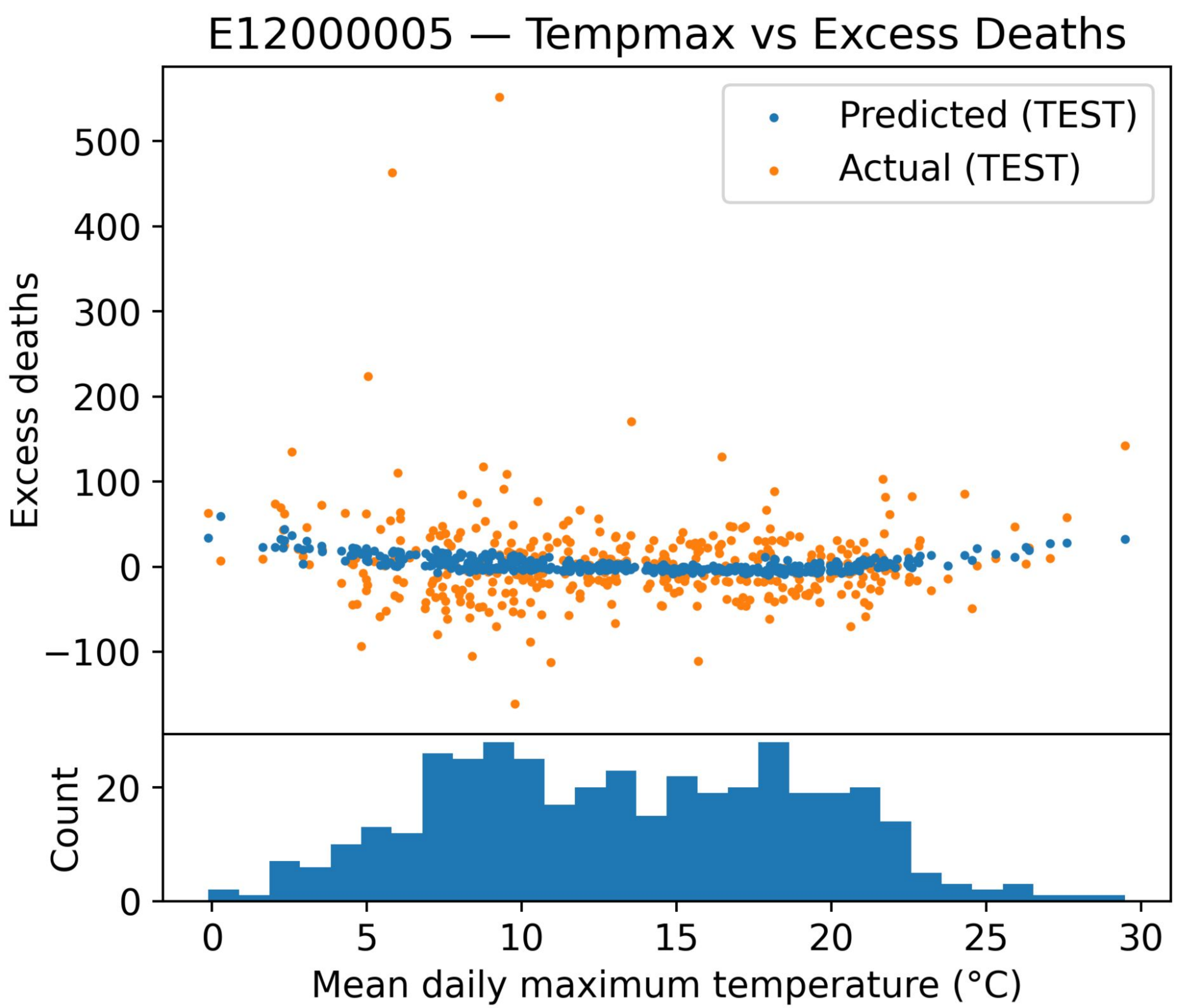
- A percentage (e.g., 81.6%) showing how much of the daily death rate is "explained" by the model (temperature, season, and time).

MAE (Mean Absolute Error)

- The average error of the model. An MAE of 12 means the model's prediction was off by an average of 12 deaths per day.

A CNN model

The trained model takes weather data (temperature, precipitation etc.) and predicts the effect on that week's deaths.



Limitation: Random weekly fluctuations in the data are stronger than the weather effects, even if a pattern is still identifiable. The model is therefore likely to be more useful for predicting long term trends than e.g. short term effects for healthcare preparedness.

Plot loc: /Data/PyTorchModellingAndPredictions/CNN_Modelling_deseasonalised/region_splits/E12000005

Problem Recap

Aim: To quantify how weeks of unusually hot or cold weather shift mortality vs. a seasonal baseline.

- **Stakeholders**
  - Public health agencies, emergency planners, health services, local authorities, the public.

- **Decision context**
  - Short-term: gauge if forecasts of extreme temps can inform surge planning.
  - Longer-term: assess implications of climate change for public health planning and risk communication.

- **Unit of analysis**
  - Region–week: **weekly excess mortality** linked to **weekly aggregated weather** (e.g., temp, precipitation, snow).

- **KPIs**
  - Accuracy on unseen historical weeks (e.g., RMSE/MAE/R^2 vs. zero baseline).
  - Consistency of effect direction/magnitude across regions (and age groups where applicable).
  - Clarity/usefulness of stakeholder-facing summaries & indicators.
  - Reproducibility of data processing and modeling pipeline.